

```

#ifndef __calc_B__
#define __calc_B__

#ifndef __common__
#include "common.hpp"
#endif

#ifndef __pk_lin__
#include "pk_lin.hpp"
#endif

#ifndef __kernel__
#include "kernel.hpp"
#endif

#ifndef __kernel2__
#include "kernel2.hpp"
#endif

#ifndef __sph__
#include "sph.hpp"
#endif

int integrand_B_Tree(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin,
    int num_k_bin, int ell1, int ell2, int ELL,
    double kmag1,
    double alpha_perp, double alpha_parallel, double sigma8, double fz,
    double b1, double b2, double bK2) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree(kvec1, kvec2, kvec3, los, alpha_perp, alpha_p
arallel, sigma8, fz, b1, b2, bK2);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

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int integrand_B_Tree_FoG(double * xx_in, int ndim, double * ff_out, int ncomp, double * k
bin, int num_k_bin, int ell1, int ell2, int ELL,
                        double kmag1,
                        double alpha_perp, double alpha_parallel, double sigma8, double fz,
double b1, double b2, double bK2,
                        double c1, double c2, double knl) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_FoG(kvec1, kvec2, kvec3, los, alpha_perp, alp
ha_parallel, sigma8, fz, b1, b2, bK2, c1, c2, knl);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_DampIvanov(double * xx_in, int ndim, double * ff_out, int ncomp, dou
ble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                        double kmag1,
                        double alpha_perp, double alpha_parallel, double sigma8, double fz,
double b1, double b2, double bK2,
                        double rbao, double ks) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

    /*****/

    double S2 = Sig2(rbao, ks);
    double dS2 = dSig2(rbao, ks);

    for(int j = 0; j < num_k_bin; j++) {

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/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_Ivanov_Damping(kvec1, kvec2, kvec3, los, alph
a_perp, alpha_parallel, sigma8, fz, b1, b2, bK2, S2, dS2);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_FoG_Damping_diag(double * xx_in, int ndim, double * ff_out, int ncomp, do
uble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double alpha_perp, double alpha_parallel, double f,
double b1, double b2, double bG2,
double c1, double c2, double knl,
double Sigma2, double dSigma2) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kbin[j]};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_FoG_Damping(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel, f, b1, b2, bG2, c1, c2, knl, Sigma2, dSigma2);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

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        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_FoG_Damping(double * xx_in, int ndim, double * ff_out, int ncomp, double
* kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1,
    double alpha_perp, double alpha_parallel, double f,
    double b1, double b2, double bG2,
    double c1, double c2, double knl,
    double Sigma2, double dSigma2) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_Tree_FoG_Damping(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel, f, b1, b2, bG2, c1, c2, knl, Sigma2, dSigma2);
        /***/

        /***/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /***/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_SN_FoG_Damping_diag(double * xx_in, int ndim, double * ff_out, int ncomp,
    double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
    double alpha_perp, double alpha_parallel, double f,
    double b1,
    double c1, double c2, double knl, double Pshot, double Bs
hot,
    double Sigma2, double dSigma2) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

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    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kbin[j]};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_SN_FoG_Damping(kvec1, kvec2, kvec3, los, alpha_perp, a
lpha_parallel, f, b1, c1, c2, knl, Pshot, Bshot, Sigma2, dSigma2);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_SN_FoG_Damping(double * xx_in, int ndim, double * ff_out, int ncomp, doub
le * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1,
    double alpha_perp, double alpha_parallel, double f,
    double b1,
    double c1, double c2, double knl, double Pshot, double Bs
hot,
    double Sigma2, double dSigma2) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_SN_FoG_Damping(kvec1, kvec2, kvec3, los, alpha_perp, a
lpha_parallel, f, b1, c1, c2, knl, Pshot, Bshot, Sigma2, dSigma2);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

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        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_PNG_Damping_diag(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double alpha_perp, double alpha_parallel, double f, double b1, double fnlloc, double fnlequi, double fnlortho, double Sigma2, double dSigma2) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kbin[j]};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_PNG_Damping(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel, f, b1, fnlloc, fnlequi, fnlortho, Sigma2, dSigma2);
        /***/

        /***/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /***/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

        ff_out[j] = result * jacobian;

    }
    return 0;
}

double Bispectrum_Kernel(double * kvec1, double * kvec2, double * los, double f, char * parameters_in) {

    std::string parameters = parameters_in;

    if(0) {

    } else if (parameters == "b1_b1_b1") {
        return Bispectrum_Kernel_b1_b1_b1(kvec1, kvec2);
    } else if (parameters == "b1_b1_b2") {
        return Bispectrum_Kernel_b1_b1_b2(kvec1, kvec2);
    } else if (parameters == "b1_b1_bG2") {
        return Bispectrum_Kernel_b1_b1_bG2(kvec1, kvec2);
    } else if (parameters == "b1_b1_f") {

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```
    return Bispectrum_Kernel_b1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_b1_b1_f") {
    return Bispectrum_Kernel_b1_b1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_b1_f_f") {
    return Bispectrum_Kernel_b1_b1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_b2_f") {
    return Bispectrum_Kernel_b1_b2_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_bG2_f") {
    return Bispectrum_Kernel_b1_bG2_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_f_f") {
    return Bispectrum_Kernel_b1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "b1_f_f_f") {
    return Bispectrum_Kernel_b1_f_f_f(kvec1, kvec2, los, f);
} else if (parameters == "b2_f_f") {
    return Bispectrum_Kernel_b2_f_f(kvec1, kvec2, los, f);
} else if (parameters == "bG2_f_f") {
    return Bispectrum_Kernel_bG2_f_f(kvec1, kvec2, los, f);
} else if (parameters == "f_f_f") {
    return Bispectrum_Kernel_f_f_f(kvec1, kvec2, los, f);
} else if (parameters == "f_f_f_f") {
    return Bispectrum_Kernel_f_f_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_b1_b1") {
    return Bispectrum_Kernel_c1_b1_b1(kvec1, kvec2, los);
} else if (parameters == "c1_b1_b2") {
    return Bispectrum_Kernel_c1_b1_b2(kvec1, kvec2, los);
} else if (parameters == "c1_b1_bG2") {
    return Bispectrum_Kernel_c1_b1_bG2(kvec1, kvec2, los);
} else if (parameters == "c1_b1_f") {
    return Bispectrum_Kernel_c1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_b1_b1_f") {
    return Bispectrum_Kernel_c1_b1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_b1_f_f") {
    return Bispectrum_Kernel_c1_b1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_b2_f") {
    return Bispectrum_Kernel_c1_b2_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_bG2_f") {
    return Bispectrum_Kernel_c1_bG2_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_f_f") {
    return Bispectrum_Kernel_c1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_f_f_f") {
    return Bispectrum_Kernel_c1_f_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_c1_b1") {
    return Bispectrum_Kernel_c1_c1_b1(kvec1, kvec2, los);
} else if (parameters == "c1_c1_b2") {
    return Bispectrum_Kernel_c1_c1_b2(kvec1, kvec2, los);
} else if (parameters == "c1_c1_bG2") {
    return Bispectrum_Kernel_c1_c1_bG2(kvec1, kvec2, los);
} else if (parameters == "c1_c1_f") {
    return Bispectrum_Kernel_c1_c1_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_c1_b1_f") {
    return Bispectrum_Kernel_c1_c1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c1_c1_f_f") {
    return Bispectrum_Kernel_c1_c1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_b1_b1") {
    return Bispectrum_Kernel_c2_b1_b1(kvec1, kvec2, los);
} else if (parameters == "c2_b1_b2") {
    return Bispectrum_Kernel_c2_b1_b2(kvec1, kvec2, los);
} else if (parameters == "c2_b1_bG2") {
    return Bispectrum_Kernel_c2_b1_bG2(kvec1, kvec2, los);
} else if (parameters == "c2_b1_f") {
    return Bispectrum_Kernel_c2_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_b1_b1_f") {
    return Bispectrum_Kernel_c2_b1_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_b1_f_f") {
    return Bispectrum_Kernel_c2_b1_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_b2_f") {
    return Bispectrum_Kernel_c2_b2_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_bG2_f") {
    return Bispectrum_Kernel_c2_bG2_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_f_f") {
    return Bispectrum_Kernel_c2_f_f(kvec1, kvec2, los, f);
}
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    return Bispectrum_Kernel_c2_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_f_f_f") {
    return Bispectrum_Kernel_c2_f_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c1_b1") {
    return Bispectrum_Kernel_c1_c2_b1(kvec1, kvec2, los);
} else if (parameters == "c2_c1_b2") {
    return Bispectrum_Kernel_c1_c2_b2(kvec1, kvec2, los);
} else if (parameters == "c2_c1_bG2") {
    return Bispectrum_Kernel_c1_c2_bG2(kvec1, kvec2, los);
} else if (parameters == "c2_c1_f") {
    return Bispectrum_Kernel_c1_c2_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c1_b1_f") {
    return Bispectrum_Kernel_c1_c2_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c1_f_f") {
    return Bispectrum_Kernel_c1_c2_f_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c2_b1") {
    return Bispectrum_Kernel_c2_c2_b1(kvec1, kvec2, los);
} else if (parameters == "c2_c2_b2") {
    return Bispectrum_Kernel_c2_c2_b2(kvec1, kvec2, los);
} else if (parameters == "c2_c2_bG2") {
    return Bispectrum_Kernel_c2_c2_bG2(kvec1, kvec2, los);
} else if (parameters == "c2_c2_f") {
    return Bispectrum_Kernel_c2_c2_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c2_b1_f") {
    return Bispectrum_Kernel_c2_c2_b1_f(kvec1, kvec2, los, f);
} else if (parameters == "c2_c2_f_f") {
    return Bispectrum_Kernel_c2_c2_f_f(kvec1, kvec2, los, f);
} else {
    return 0.0;
}
}

int integrand_B_Kernel(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin,
int num_k_bin, int ell1, int ell2, int ELL,
double kmag1, double f,
double Sigma2, double dSigma2, char * parameters) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double K12 = Bispectrum_Kernel(kvec1, kvec2, los, f, parameters);
        double K13 = Bispectrum_Kernel(kvec1, kvec3, los, f, parameters);
        double K23 = Bispectrum_Kernel(kvec2, kvec3, los, f, parameters);
        /***/

        double k1 = NORM(kvec1);
        double k2 = NORM(kvec2);
        double k3 = NORM(kvec3);

        double BA01 = f_pk(k1) - f_pk_no_wiggle(k1);
        double BA02 = f_pk(k2) - f_pk_no_wiggle(k2);
        double BA03 = f_pk(k3) - f_pk_no_wiggle(k3);
    }
}

```

```

        double bispec = K12 * (BAO1 * BAO2
                                * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma2)
                                + BAO1 * f_pk_no_wiggle(k2)
                                * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2)
                                + f_pk_no_wiggle(k1) * BAO2
                                * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2)
                                + f_pk_no_wiggle(k1) * f_pk_no_wiggle(k2))
        + K13 * (BAO1 * BAO3
                                * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma2)
                                + BAO1 * f_pk_no_wiggle(k3)
                                * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2)
                                + f_pk(k1) * BAO3
                                * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma
2)
                                + f_pk_no_wiggle(k1) * f_pk_no_wiggle(k3))
        + K23 * (BAO2 * BAO3
                                * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma2)
                                + BAO2 * f_pk_no_wiggle(k3)
                                * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2)
                                + f_pk(k2) * BAO3
                                * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma
2)
                                + f_pk_no_wiggle(k2) * f_pk_no_wiggle(k3));

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Kernel_diag(double * xx_in, int ndim, double * ff_out, int ncomp, double
* kbin, int num_k_bin, int ell1, int ell2, int ELL,
                        double f,
                        double Sigma2, double dSigma2, double alpha_perp, double alph
a_parallel, char * parameters) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1_in[3] = {0.0, 0.0, kbin[j]};
        double kvec2_in[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3_in[3] = {- kvec1_in[0] - kvec2_in[0], - kvec1_in[1] - kvec2_in[1
], - kvec1_in[2] - kvec2_in[2]};

```

```

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

double kvec1[3] = { 0.0, 0.0, 0.0 };
double kvec2[3] = { 0.0, 0.0, 0.0 };
double kvec3[3] = { 0.0, 0.0, 0.0 };

if (alpha_parallel == 0){
    kvec1[0] = kvec1_in[0];
    kvec1[1] = kvec1_in[1];
    kvec1[2] = kvec1_in[2];
    kvec2[0] = kvec2_in[0];
    kvec2[1] = kvec2_in[1];
    kvec2[2] = kvec2_in[2];
    kvec3[0] = kvec3_in[0];
    kvec3[1] = kvec3_in[1];
    kvec3[2] = kvec3_in[2];
}
else {
    calcTrueWavevector(kvec1_in, los, alpha_perp, alpha_parallel, kvec1);
    calcTrueWavevector(kvec2_in, los, alpha_perp, alpha_parallel, kvec2);
    calcTrueWavevector(kvec3_in, los, alpha_perp, alpha_parallel, kvec3);
}

/*****/
double K12 = Bispectrum_Kernel(kvec1, kvec2, los, f, parameters);
double K13 = Bispectrum_Kernel(kvec1, kvec3, los, f, parameters);
double K23 = Bispectrum_Kernel(kvec2, kvec3, los, f, parameters);
/*****/

double k1 = NORM(kvec1);
double k2 = NORM(kvec2);
double k3 = NORM(kvec3);

double BAO1 = f_pk(k1) - f_pk_no_wiggle(k1);
double BAO2 = f_pk(k2) - f_pk_no_wiggle(k2);
double BAO3 = f_pk(k3) - f_pk_no_wiggle(k3);

double bispec = K12 * (BAO1 * BAO2
    * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma2)
    + BAO1 * f_pk_no_wiggle(k2)
    * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2)
    + f_pk_no_wiggle(k1) * BAO2
    * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2)
    + f_pk_no_wiggle(k1) * f_pk_no_wiggle(k2))
    + K13 * (BAO1 * BAO3
    * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma2)
    + BAO1 * f_pk_no_wiggle(k3)
    * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma
2)
    + f_pk(k1) * BAO3
    * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma
2)
    + f_pk_no_wiggle(k1) * f_pk_no_wiggle(k3))
    + K23 * (BAO2 * BAO3
    * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2) * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma2)
    + BAO2 * f_pk_no_wiggle(k3)
    * ExpDamping_Ivanov(kvec2, los, f, Sigma2, dSigma
2)
    + f_pk(k2) * BAO3
    * ExpDamping_Ivanov(kvec3, los, f, Sigma2, dSigma
2)
    + f_pk_no_wiggle(k2) * f_pk_no_wiggle(k3));

```

```

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

    ff_out[j] = result * jacobian;

}
return 0;
}

double Bispectrum_Kernel_SN(double * kvec1, double * los, double f, char * parameters_in)
{
    std::string parameters = parameters_in;

    if(0) {

    } else if (parameters == "Bshot_b1_b1") {
        return Bispectrum_Kernel_Bshot_b1_b1();
    } else if (parameters == "Bshot_b1_f") {
        return Bispectrum_Kernel_Bshot_b1_f(kvec1, los, f);
    } else if (parameters == "Bshot_b1_c1") {
        return Bispectrum_Kernel_Bshot_b1_c1(kvec1, los);
    } else if (parameters == "Bshot_b1_c2") {
        return Bispectrum_Kernel_Bshot_b1_c2(kvec1, los);
    } else if (parameters == "Pshot_f_b1") {
        return Bispectrum_Kernel_Pshot_f_b1(kvec1, los, f);
    } else if (parameters == "Pshot_f_f") {
        return Bispectrum_Kernel_Pshot_f_f(kvec1, los, f);
    } else if (parameters == "Pshot_f_c1") {
        return Bispectrum_Kernel_Pshot_f_c1(kvec1, los, f);
    } else if (parameters == "Pshot_f_c2") {
        return Bispectrum_Kernel_Pshot_f_c2(kvec1, los, f);
    } else {
        return 0.0;
    }
}

int integrand_B_Kernel_SN(double * xx_in, int ndim, double * ff_out, int ncomp, double *
kbin, int num_k_bin, int ell1, int ell2, int ELL,
                        double kmag1, double f,
                        double Sigma2, double dSigma2, char * parameters) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/

```

```

double K1 = Bispectrum_Kernel_SN(kvec1, los, f, parameters);
double K2 = Bispectrum_Kernel_SN(kvec2, los, f, parameters);
double K3 = Bispectrum_Kernel_SN(kvec3, los, f, parameters);
/*****/

double k1 = NORM(kvec1);
double k2 = NORM(kvec2);
double k3 = NORM(kvec3);

double BA01 = f_pk(k1) - f_pk_no_wiggle(k1);
double BA02 = f_pk(k2) - f_pk_no_wiggle(k2);
double BA03 = f_pk(k3) - f_pk_no_wiggle(k3);

double bispec = K1 * (BA01 * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma2
) + f_pk_no_wiggle(k1))
+ K2 * (BA02 * ExpDamping_Ivanov(kvec2, los, f, Sigma2, d
Sigma2) + f_pk_no_wiggle(k2))
+ K3 * (BA03 * ExpDamping_Ivanov(kvec3, los, f, Sigma2, d
Sigma2) + f_pk_no_wiggle(k3));

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Kernel_SN_diag(double * xx_in, int ndim, double * ff_out, int ncomp, doub
le * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double f,
double Sigma2, double dSigma2, double alpha_perp, double alph
a_parallel, char * parameters) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1_in[3] = {0.0, 0.0, kbin[j]};
double kvec2_in[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3_in[3] = {- kvec1_in[0] - kvec2_in[0], - kvec1_in[1] - kvec2_in[1
], - kvec1_in[2] - kvec2_in
[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};

double kvec1[3] = { 0.0, 0.0, 0.0 };
double kvec2[3] = { 0.0, 0.0, 0.0 };
double kvec3[3] = { 0.0, 0.0, 0.0 };
/*****/

if (alpha_parallel == 0){
kvec1[0] = kvec1_in[0];
kvec1[1] = kvec1_in[1];
kvec1[2] = kvec1_in[2];

```

```

        kvec2[0] = kvec2_in[0];
        kvec2[1] = kvec2_in[1];
        kvec2[2] = kvec2_in[2];
        kvec3[0] = kvec3_in[0];
        kvec3[1] = kvec3_in[1];
        kvec3[2] = kvec3_in[2];
    }
    else {
        calcTrueWavevector(kvec1_in, los, alpha_perp, alpha_parallel, kvec1);
        calcTrueWavevector(kvec2_in, los, alpha_perp, alpha_parallel, kvec2);
        calcTrueWavevector(kvec3_in, los, alpha_perp, alpha_parallel, kvec3);
    }

    /*****/
    double K1 = Bispectrum_Kernel_SN(kvec1, los, f, parameters);
    double K2 = Bispectrum_Kernel_SN(kvec2, los, f, parameters);
    double K3 = Bispectrum_Kernel_SN(kvec3, los, f, parameters);
    /*****/

    double k1 = NORM(kvec1);
    double k2 = NORM(kvec2);
    double k3 = NORM(kvec3);

    double BAO1 = f_pk(k1) - f_pk_no_wiggle(k1);
    double BAO2 = f_pk(k2) - f_pk_no_wiggle(k2);
    double BAO3 = f_pk(k3) - f_pk_no_wiggle(k3);

    double bispec = K1 * (BAO1 * ExpDamping_Ivanov(kvec1, los, f, Sigma2, dSigma2
) + f_pk_no_wiggle(k1))
                    + K2 * (BAO2 * ExpDamping_Ivanov(kvec2, los, f, Sigma2, d
Sigma2) + f_pk_no_wiggle(k2))
                    + K3 * (BAO3 * ExpDamping_Ivanov(kvec3, los, f, Sigma2, d
Sigma2) + f_pk_no_wiggle(k3));

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

    ff_out[j] = result * jacobian;

}
return 0;
}

```

```

double Bispectrum_Kernel_PNG(double * kvec1, double * kvec2, double * kvec3, double * los
, double f, double Sigma2, double dSigma2, char * parameters_in) {

```

```

    std::string parameters = parameters_in;

    if(0) {

    } else if (parameters == "fnlloc_b1_b1_b1") {
        return Bispectrum_Kernel_PNG_fnlloc_b1_b1_b1(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlloc_b1_b1_f") {
        return Bispectrum_Kernel_PNG_fnlloc_b1_b1_f(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlloc_b1_f_f") {
        return Bispectrum_Kernel_PNG_fnlloc_b1_f_f(kvec1, kvec2, kvec3, los, f, Sigma2, d
Sigma2);
    } else if (parameters == "fnlloc_f_f_f") {
        return Bispectrum_Kernel_PNG_fnlloc_f_f_f(kvec1, kvec2, kvec3, los, f, Sigma2, dS
igma2);
    } else if (parameters == "fnlequi_b1_b1_b1") {
        return Bispectrum_Kernel_PNG_fnlequi_b1_b1_b1(kvec1, kvec2, kvec3, los, f, Sigma2

```

```
, dSigma2);
    } else if (parameters == "fnlequi_b1_b1_f") {
        return Bispectrum_Kernel_PNG_fnlequi_b1_b1_f(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlequi_b1_f_f") {
        return Bispectrum_Kernel_PNG_fnlequi_b1_f_f(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlequi_f_f_f") {
        return Bispectrum_Kernel_PNG_fnlequi_f_f_f(kvec1, kvec2, kvec3, los, f, Sigma2, d
Sigma2);
    } else if (parameters == "fnlortho_b1_b1_b1") {
        return Bispectrum_Kernel_PNG_fnlortho_b1_b1_b1(kvec1, kvec2, kvec3, los, f, Sigma
2, dSigma2);
    } else if (parameters == "fnlortho_b1_b1_f") {
        return Bispectrum_Kernel_PNG_fnlortho_b1_b1_f(kvec1, kvec2, kvec3, los, f, Sigma2
, dSigma2);
    } else if (parameters == "fnlortho_b1_f_f") {
        return Bispectrum_Kernel_PNG_fnlortho_b1_f_f(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlortho_f_f_f") {
        return Bispectrum_Kernel_PNG_fnlortho_f_f_f(kvec1, kvec2, kvec3, los, f, Sigma2,
dSigma2);
    } else if (parameters == "fnlortho_LSS_b1_b1_b1") {
        return Bispectrum_Kernel_PNG_fnlortho_LSS_b1_b1_b1(kvec1, kvec2, kvec3, los, f, S
igma2, dSigma2);
    } else if (parameters == "fnlortho_LSS_b1_b1_f") {
        return Bispectrum_Kernel_PNG_fnlortho_LSS_b1_b1_f(kvec1, kvec2, kvec3, los, f, Si
gma2, dSigma2);
    } else if (parameters == "fnlortho_LSS_b1_f_f") {
        return Bispectrum_Kernel_PNG_fnlortho_LSS_b1_f_f(kvec1, kvec2, kvec3, los, f, Sig
ma2, dSigma2);
    } else if (parameters == "fnlortho_LSS_f_f_f") {
        return Bispectrum_Kernel_PNG_fnlortho_LSS_f_f_f(kvec1, kvec2, kvec3, los, f, Sigm
a2, dSigma2);
    } else {
        return 0.0;
    }
}
```

```
int integrand_B_Kernel_PNG_diag(double * xx_in, int ndim, double * ff_out, int ncomp, dou
ble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double f,
double Sigma2, double dSigma2, char * parameters) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kbin[j]};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Kernel_PNG(kvec1, kvec2, kvec3, los, f, Sigma2, dS
igma2, parameters);
        /*****/

        /*****/
    }
}
```

```

double Nl1l = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double Sl1l = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl1l * Sl1l * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_NoWiggle(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma8, double fz, double b1, double b2, double bK2) {

```

```

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel, sigma8, fz, b1, b2, bK2);
/*****/

/*****/
double Nl1l = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double Sl1l = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl1l * Sl1l * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_NoWiggle_RealSpace_DarkMatter_Growth(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double sigma8) {

/*****/
double mu2 = - 1.0 + 2.0 * xx_in[0];
/*****/

```

```

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_RealSpace_DarkMatter_Growth(kvec1, k
vec2, kvec3, sigma8);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_NoWiggle_RealSpace_DarkMatter_Shift(double * xx_in, int ndim, double
* ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                double kmag1,
                double sigma8) {

    /*****/
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_RealSpace_DarkMatter_Shift(kvec1, kv
ec2, kvec3, sigma8);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_NoWiggle_RealSpace_DarkMatter_Tidal(double * xx_in, int ndim, double
* ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                double kmag1,
```

```

        double sigma8) {

    /*******/
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*******/

        /*******/
        double bispec = Bispectrum_Tree_NoWiggle_RealSpace_DarkMatter_Tidal(kvec1, kv
ec2, kvec3, sigma8);
        /*******/

        /*******/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*******/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_BAO_RealSpace_DarkMatter_Growth(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double sigma8, double sigma2_perp) {

    /*******/
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*******/

        /*******/
        double bispec = Bispectrum_Tree_BAO_RealSpace_DarkMatter_Growth(kvec1, kvec2,
kvec3, sigma8, sigma2_perp);
        /*******/

        /*******/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*******/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

```

```

}
```

```

int integrand_B_Tree_BAO_RealSpace_DarkMatter_Shift(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double sigma8, double sigma2_perp) {

    /*****/
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_RealSpace_DarkMatter_Shift(kvec1, kvec2,
kvec3, sigma8, sigma2_perp);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_BAO_RealSpace_DarkMatter_Tidal(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double sigma8, double sigma2_perp) {

    /*****/
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_RealSpace_DarkMatter_Tidal(kvec1, kvec2,
kvec3, sigma8, sigma2_perp);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0);
        double Sl11 = LegendrePolynomials(ell1, mu2);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
```

```

        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_BAO(double * xx_in, int ndim, double * ff_out, int ncomp, double * k
bin, int num_k_bin, int ell1, int ell2, int ELL,
                        double kmag1,
                        double alpha_perp, double alpha_parallel, double sigma8, double
fz, double b1, double b2, double bK2, double sigma2_perp, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO(kvec1, kvec2, kvec3, los, alpha_perp, alp
ha_parallel, sigma8, fz, b1, b2, bK2, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_Reconstructed(double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                        double kmag1,
                        double alpha_perp, double alpha_parallel, double sigma
8, double fz, double b1, double b2, double bK2, double b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

```

```

    /***/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /***/

    /***/
    double bispec = Bispectrum_Tree_Reconstructed(kvec1, kvec2, kvec3, los, alpha
_perp, alpha_parallel, sigma8, fz, b1, b2, bK2, b1_fid, R);
    /***/

    /***/
    double Nlll = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Slll = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nlll * Slll * bispec;
    /***/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_Tree_NoWiggle_Reconstructed(double * xx_in, int ndim, double * ff_out, in
t ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, dou
ble sigma8, double fz, double b1, double b2, double bK2, double b1_fid, double R) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_Tree_NoWiggle_Reconstructed(kvec1, kvec2, kvec3, l
os, alpha_perp, alpha_parallel, sigma8, fz, b1, b2, bK2, b1_fid, R);
        /***/

        /***/
        double Nlll = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Slll = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nlll * Slll * bispec;
        /***/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));

```

```

        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_Local(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                   double kmag1,
                                   double alpha_perp, double alpha_parallel, double sigma8
, double fz, double b1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Local(kvec1, kvec2, kvec3, los, alpha_
perp, alpha_parallel, sigma8, fz, b1);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_Equilateral(double * xx_in, int ndim, double * ff_out, int nc
omp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                   double kmag1,
                                   double alpha_perp, double alpha_parallel, double
sigma8, double fz, double b1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

```

```

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_Equilateral(kvec1, kvec2, kvec3, los,
alpha_perp, alpha_parallel, sigma8, fz, b1);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_Orthogonal(double * xx_in, int ndim, double * ff_out, int nco
mp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double s
igma8, double fz, double b1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Orthogonal(kvec1, kvec2, kvec3, los, a
lpha_perp, alpha_parallel, sigma8, fz, b1);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;
    }
}

```

```

    }
    return 0;
}

int integrand_B_Tree_b1_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*******/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*******/
        double bispec = Bispectrum_Tree_b1_b1_b1(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel);
        /*******/

        /*******/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*******/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }

    return 0;
}

int integrand_B_Tree_b1_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*******/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -

```

```

kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_b1_b1_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_b1_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
* kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_b1_f_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_b2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                double kmag1,
                                double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_b2_b1_b1(kvec1, kvec2, kvec3, los, alpha_perp
, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_b2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
* kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                double kmag1,
                                double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/

```

```

    double bispec = Bispectrum_Tree_b2_b1_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_b2_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
* kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_b2_f_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

/*****/

int integrand_B_Tree_bK2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, doub
le * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/

```

```

double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_bK2_b1_b1(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell11)+1.0) * (2.0*double(ell12)+1.0) * (2.0*double(ELL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell11, ell12, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_bK2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell11, int ell12, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_bK2_b1_f(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell11)+1.0) * (2.0*double(ell12)+1.0) * (2.0*double(ELL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell11, ell12, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
}

```

```

LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /***/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_bK2_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
    * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_Tree_bK2_f_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
        /***/

        /***/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /***/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

/***/

int integrand_B_Tree_b1f_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, doub
le * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];

```

```

double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_b1f_b1_b1(kvec1, kvec2, kvec3, los, alpha_per
p, alpha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_b1f_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, doubl
e * kbin, int num_k_bin, int ell1, int ell2, int ELL,
    double kmag1,
    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_b1f_b1_f(kvec1, kvec2, kvec3, los, alpha_perp
, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

```

```

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_b1f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
    * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_b1f_f_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

/*****/

int integrand_B_Tree_ff_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double
    * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

```

```

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_ff_f_f(kvec1, kvec2, kvec3, los, alpha_perp,
alpha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double *
kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_f_f_f(kvec1, kvec2, kvec3, los, alpha_perp, a
lpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
}

```

```

    return 0;
}

int integrand_B_Tree_Reconstructed_b1b1_b1_b1(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, d
ouble b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_Reconstructed_b1b1_b1_b1(kvec1, kvec2, kvec3,
los, alpha_perp, alpha_parallel, b1_fid, R);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_Reconstructed_b1b1_b1_f(double * xx_in, int ndim, double * ff_out, i
nt ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};

```

```

double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_Reconstructed_b1b1_b1_f(kvec1, kvec2, kvec3,
los, alpha_perp, alpha_parallel, b1_fid, R);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_Reconstructed_b1b1_f_f(double * xx_in, int ndim, double * ff_out, in
t ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_Reconstructed_b1b1_f_f(kvec1, kvec2, kvec3, l
os, alpha_perp, alpha_parallel, b1_fid, R);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;

```

```

}

int integrand_B_Tree_Reconstructed_blf_f_f(double * xx_in, int ndim, double * ff_out, int
ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble bl_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_Reconstructed_blf_f_f(kvec1, kvec2, kvec3, lo
s, alpha_perp, alpha_parallel, bl_fid, R);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_Reconstructed_ff_f_f(double * xx_in, int ndim, double * ff_out, int
ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble bl_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};

```

```

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_Reconstructed_ff_f_f(kvec1, kvec2, kvec3, los
, alpha_perp, alpha_parallel, b1_fid, R);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

/*****/
/** NoWiggle **/
/*****/

int integrand_B_Tree_NoWiggle_b1_b1_b1(
double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1,
double alpha_perp, double alpha_parallel
) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/
double bispec = Bispectrum_Tree_NoWiggle_b1_b1_b1(kvec1, kvec2, kvec3, los, a
lpha_perp, alpha_parallel);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}

return 0;
}

```

```

int integrand_B_Tree_NoWiggle_b1_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_b1_b1_f(kvec1, kvec2, kvec3, los, al
pha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_NoWiggle_b1_f_f(double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

```

```

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_b1_f_f(kvec1, kvec2, kvec3, los, alp
ha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_NoWiggle_b2_b1_b1(double * xx_in, int ndim, double * ff_out, int nco
mp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_b2_b1_b1(kvec1, kvec2, kvec3, los, a
lpha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_NoWiggle_b2_b1_f(double * xx_in, int ndim, double * ff_out, int ncom
p, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
        double kmag1,
        double alpha_perp, double alpha_parallel) {

```

```

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_Tree_NoWiggle_b2_b1_f(kvec1, kvec2, kvec3, los, al
pha_perp, alpha_parallel);
        /***/

        /***/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /***/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_NoWiggle_b2_f_f(double * xx_in, int ndim, double * ff_out, int ncomp
, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
    double kmag1,
    double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double bispec = Bispectrum_Tree_NoWiggle_b2_f_f(kvec1, kvec2, kvec3, los, alp
ha_perp, alpha_parallel);
        /***/

        /***/

```

```

double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

/*****/

int integrand_B_Tree_NoWiggle_bK2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_bK2_b1_b1(kvec1, kvec2, kvec3, los,
alpha_perp, alpha_parallel);
/*****/

/*****/
double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_NoWiggle_bK2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];

```

```

double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_NoWiggle_bK2_b1_f(kvec1, kvec2, kvec3, los, a
lpha_perp, alpha_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_NoWiggle_bK2_f_f(double * xx_in, int ndim, double * ff_out, int ncom
p, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_bK2_f_f(kvec1, kvec2, kvec3, los, al
pha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

```

```

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

/*****/

int integrand_B_Tree_NoWiggle_b1f_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_b1f_b1_b1(kvec1, kvec2, kvec3, los,
alpha_perp, alpha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_NoWiggle_b1f_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

```

```

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_b1f_b1_f(kvec1, kvec2, kvec3, los, a
lpha_perp, alpha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_NoWiggle_b1f_f_f(double * xx_in, int ndim, double * ff_out, int ncom
p, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_b1f_f_f(kvec1, kvec2, kvec3, los, al
pha_perp, alpha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;

```

```

}

/*****/

int integrand_B_Tree_NoWiggle_ff_f_f(double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_ff_f_f(kvec1, kvec2, kvec3, los, alp
ha_perp, alpha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_NoWiggle_f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp,
double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -

```

```

    kvec2[2]};
    double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_NoWiggle_f_f_f(kvec1, kvec2, kvec3, los, alph
a_perp, alpha_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_NoWiggle_Reconstructed_b1b1_b1_b1(double * xx_in, int ndim, double *
ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, d
ouble b1_fid, double R) {

    /*****/
    double mu1    = 1.0;
    double phi1   = 0.0;
    double mu2    = - 1.0 + 2.0 * xx_in[0];
    double phi2   = 0.0;
    double mu     = - 1.0 + 2.0 * xx_in[1];
    double phi    = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_Reconstructed_b1b1_b1_b1(kvec1, kvec
2, kvec3, los, alpha_perp, alpha_parallel, b1_fid, R);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_NoWiggle_Reconstructed_b1b1_b1_f(double * xx_in, int ndim, double *
ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_Reconstructed_b1b1_b1_f(kvec1, kvec2
, kvec3, los, alpha_perp, alpha_parallel, b1_fid, R);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_NoWiggle_Reconstructed_b1b1_f_f(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};

```

```

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_Reconstructed_b1b1_f_f(kvec1, kvec2,
kvec3, los, alpha_perp, alpha_parallel, b1_fid, R);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_NoWiggle_Reconstructed_b1f_f_f(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_NoWiggle_Reconstructed_b1f_f_f(kvec1, kvec2,
kvec3, los, alpha_perp, alpha_parallel, b1_fid, R);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

```

```

int integrand_B_Tree_NoWiggle_Reconstructed_ff_f_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, do
uble b1_fid, double R) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_NoWiggle_Reconstructed_ff_f_f(kvec1, kvec2, k
vec3, los, alpha_perp, alpha_parallel, b1_fid, R);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

//int integrand_B_DDD_SS(double * xx_in, int ndim, double * ff_out, int ncomp, double * k
bin, int num_k_bin,
//          int ell1, int ell2, int ELL, int ell1_dash, int ell2_dash, int ELL
_dash,
//          double kmag1,
//          double alpha_perp, double alpha_parallel) {
//
//      /*****/
//      double mu1 = 1.0;
//      double phi1 = 0.0;
//      double mu2 = - 1.0 + 2.0 * xx_in[0];
//      double phi2 = 0.0;
//      double mu = - 1.0 + 2.0 * xx_in[1];
//      double phi = 0.0 + 2.0 * M_PI * xx_in[2];
//      /*****/
//
//      for(int j = 0; j < num_k_bin; j++) {
//
//          /*****/
//          double kvec1[3] = {0.0, 0.0, kmag1};
//          double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
//          double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
//
//          double kvec1_dash[3] = { 0.0, 0.0, 0.0, };
//          double kvec2_dash[3] = { 0.0, 0.0, 0.0, };
//          /*****/
//
//          /*****/
//          calcTrueWavevector(kvec1, los, alpha_perp, alpha_parallel, kvec1_dash);
//          calcTrueWavevector(kvec2, los, alpha_perp, alpha_parallel, kvec2_dash);
//          /*****/
//
//          /*****/
//          double YYY_YYY = calcYYY_YYY(kvec1, kvec2, kvec1_dash, kvec2_dash, los, ell1,
ell2, ELL, ell1_dash, ell2_dash, ELL_dash);
//          /*****/
//
//          /*****/
//          double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
//          double result = N111 * YYY_YYY;
//          /*****/
//
//          double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
//          ff_out[j] = result * jacobian;
//
//      }
//
//      return 0;
//}

double get_nm(int n) {

    if(0) {

```

```

    } else if (n == 0) {
        return 1.0;
    } else if (n == 1) {
        return 1.0;
    } else if (n == 2) {
        return 2.0;
    } else if (n == 3) {
        return 6.0;
    } else if (n == 4) {
        return 24.0;
    } else {
        return 1.0e-10;
    }
}

int integrand_SS(
    double * xx_in, int ndim, double * ff_out, int ncomp,
    int ell1, int ell2, int ELL, int ell1_dash, int ell2_dash, int ELL_dash,
    int n, int m,
    double * epsilon, int num_epsilon) {

    /***/
    // double mu1 = 1.0;
    // double phi1 = 0.0;
    double mu1 = - 1.0 + 2.0 * xx_in[0];
    double phi1 = 0.0 + 2.0 * M_PI * xx_in[1];
    double mu2 = - 1.0 + 2.0 * xx_in[2];
    // double phi2 = 0.0 + 2.0 * M_PI * xx_in[3];
    double phi2 = 0.0;
    /***/

    for(int i = 0; i < num_epsilon; i++) {

        /***/
        // double kvec1_hat[3] = {0.0, 0.0, 1.0};
        // double kvec2_hat[3] = {sqrt(1.0 - mu2 * mu2), 0.0, mu2};
        // double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi)
        , mu};
        double kvec1_hat[3] = {sqrt(1.0 - mu1 * mu1) * cos(phi1), sqrt(1.0 - mu1 * mu1) *
            sin(phi1), mu1};
        double kvec2_hat[3] = {sqrt(1.0 - mu2 * mu2) * cos(phi2), sqrt(1.0 - mu2 * mu2) *
            sin(phi2), mu2};
        // double kvec2_hat[3] = {sqrt(1.0 - mu2 * mu2), 0.0, mu2};
        double los[3] = {0.0, 0.0, 1.0};

        double kvec1_hat_dash[3] = { 0.0, 0.0, 1.0 };
        double kvec2_hat_dash[3] = { 0.0, 0.0, 1.0 };
        calcTrueWavevectorHat(kvec1_hat, los, epsilon[i], kvec1_hat_dash);
        calcTrueWavevectorHat(kvec2_hat, los, epsilon[i], kvec2_hat_dash);
        /***/

        /***/
        double mu1_n = MU(kvec1_hat, los);
        double mu2_m = MU(kvec2_hat, los);
        double L1 = (3.0 * mu1_n * mu1_n - 1.0) / 2.0;
        double L2 = (3.0 * mu2_m * mu2_m - 1.0) / 2.0;

        double fac1_a = (2.0/3.0) * L1 * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double fac1_b = 1.0 + (1.0/3.0) * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double Delta1 = pow( 1.0 + fac1_a / fac1_b, 0.5 ) - 1.0;
        double Delta1_n = pow( Delta1, double(n));

        double fac2_a = (2.0/3.0) * L2 * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double fac2_b = 1.0 + (1.0/3.0) * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double Delta2 = pow( 1.0 + fac2_a / fac2_b, 0.5 ) - 1.0;
        double Delta2_m = pow( Delta2, double(m));
        /***/

        /***/

```

```

    double YYY_YYY = calcYYY_YYY(kvec1_hat, kvec2_hat, kvec1_hat_dash, kvec2_hat_dash
, los, ell1, ell2, ELL, ell1_dash, ell2_dash, ELL_dash);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+
1.0);
    double NN = get_nm(n);
    double MM = get_nm(m);
    double result = N111 * YYY_YYY * Delta1_n * Delta2_m / (NN * MM);
    /*****/

    double jacobian = 1.0;
    ff_out[i] = result * jacobian;

}

    return 0;

}

int integrand_SSpow(
    double * xx_in, int ndim, double * ff_out, int ncomp,
    int ELL, int ELL_dash,
    int n,
    double * epsilon, int num_epsilon) {

    /*****/
    double mu = - 1.0 + 2.0 * xx_in[0];
    double phi = 0.0 + 2.0 * M_PI * xx_in[1];
    /*****/

    for(int i = 0; i < num_epsilon; i++) {

        /*****/
        double kvec_hat[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(p
hi), mu};
        double los[3] = {0.0, 0.0, 1.0};

        double kvec_hat_dash[3] = { 0.0, 0.0, 1.0 };
        calcTrueWavevectorHat(kvec_hat, los, epsilon[i], kvec_hat_dash);
        /*****/

        /*****/
        double mu_n = MU(kvec_hat, los);
        double mu_dash = MU(kvec_hat_dash, los);
        double L = (3.0 * mu_n * mu_n - 1.0) / 2.0;

        double fac_a = (2.0/3.0) * L * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double fac_b = 1.0 + (1.0/3.0) * ( pow(1.0 + epsilon[i], - 6.0) - 1.0 );
        double Delta = pow( 1.0 + fac_a / fac_b, 0.5 ) - 1.0;
        double Delta_n = pow( Delta, double(n));
        /*****/

        /*****/
        double YYY_YYY = LegendrePolynomials(ELL, mu_n) * LegendrePolynomials(ELL_dash, m
u_dash);
        /*****/

        /*****/
        double N111 = 2.0*double(ELL)+1.0;
        double NN = get_nm(n);
        double result = N111 * YYY_YYY * Delta_n / NN;
        /*****/

        double jacobian = 1.0;
        ff_out[i] = result * jacobian;

    }
}

```

```

    return 0;

}

/* PNG from PB */

double kmax_integ = 1.0;
//double kmax_integ = 10.0;
double kmin_integ = 1.0e-4;

int integrand_B_NonGaussian_From_PB_Local(double * xx_in, int ndim, double * ff_out, int
ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double
sigma8, double fz,
double b1, double b2, double b3, double bK2, do
uble bK3, double bDK, double bO
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local(kvec1, kvec2, kvec3, pve
c, los,
alpha_perp, alpha_parallel,
sigma8, fz,
b1, b2, b3, bK2, bK3, bDK, bO);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;
    }
}

```

```

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Equilateral(double * xx_in, int ndim, double * ff_out
, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double
sigma8, double fz,
double b1, double b2, double b3, double bK2, double bK3, double bDK, double bO
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Equilateral(kvec1, kvec2, kvec
3, pvec, los,
alpha_perp, alpha_parallel,
sigma8, fz,
b1, b2, b3, bK2, bK3, bDK, bO);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_Reconstructed(double * xx_in, int ndim, double
* ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double
sigma8, double fz,
double b1, double b2, double b3, double bK2, double bK3, double bDK, double bO, double b1_fid, double R
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_Reconstructed(kvec1, kve
c2, kvec3, pvec, los,
alpha_perp, alpha_parallel,
sigma8, fz,
b1, b2, b3, bK2, bK3, bD
K, bO, b1_fid, R);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Equilateral_Reconstructed(double * xx_in, int ndim, d
ouble * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,

```

```

double alpha_perp, double alpha_parallel, double
e sigma8, double fz,
double b1, double b2, double b3, double bK2, double
bK3, double bDK, double bO, double b1_fid, double R
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Equilateral_Reconstructed(kvec
1, kvec2, kvec3, pvec, los,
alpha_perp, alpha_parallel,
sigma8, fz,
b1, b2, b3, bK2, bK3, bDK,
bO, b1_fid, R);

        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_Tree_BAO_b1_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1,
```

```

double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {

    /*******/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*******/
        double bispec = Bispectrum_Tree_BAO_b1_b1_b1(kvec1, kvec2, kvec3, los, alpha_
perp, alpha_parallel, sigma2_perp, sigma2_para);
        /*******/

        /*******/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*******/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }

    return 0;
}

int integrand_B_Tree_BAO_b1_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, do
uble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_perp,
double sigma2_para) {

    /*******/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*******/

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*******/

        /*******/
        double bispec = Bispectrum_Tree_BAO_b1_b1_f(kvec1, kvec2, kvec3, los, alpha_p
erp, alpha_parallel, sigma2_perp, sigma2_para);

```

```

/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_b1_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, dou
ble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_perp,
double sigma2_para) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_Tree_BAO_b1_f_f(kvec1, kvec2, kvec3, los, alpha_pe
rp, alpha_parallel, sigma2_perp, sigma2_para);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_b2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, d
ouble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_per
p, double sigma2_para) {

/*****/

```

```

double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_BAO_b2_b1_b1(kvec1, kvec2, kvec3, los, alpha_
perp, alpha_parallel, sigma2_perp, sigma2_para);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_b2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, do
uble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_perp
, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_b2_b1_f(kvec1, kvec2, kvec3, los, alpha_p
erp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/

```

```

double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_b2_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {
/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {
/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
/*****/

double bispec = Bispectrum_Tree_BAO_b2_f_f(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel, sigma2_perp, sigma2_para);
/*****/

double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

/*****/

int integrand_B_Tree_BAO_bK2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {
/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];

```

```

double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_BAO_bK2_b1_b1(kvec1, kvec2, kvec3, los, alpha_
_perp, alpha_parallel, sigma2_perp, sigma2_para);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_bK2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, d
ouble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_perp
, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_bK2_b1_f(kvec1, kvec2, kvec3, los, alpha_
_perp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);

```

```

double result = N111 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_bK2_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_bK2_f_f(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

/*****/

int integrand_B_Tree_BAO_b1f_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];

```

```

double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_Tree_BAO_b1f_b1_b1(kvec1, kvec2, kvec3, los, alpha_
_perp, alpha_parallel, sigma2_perp, sigma2_para);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_Tree_BAO_b1f_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, d
ouble * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel, double sigma2_perp
, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_b1f_b1_f(kvec1, kvec2, kvec3, los, alpha_
_perp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

```

```

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_Tree_BAO_b1f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_b1f_f_f(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

/*****/

int integrand_B_Tree_BAO_ff_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

```

```

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_ff_f_f(kvec1, kvec2, kvec3, los, alpha_pe
rp, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_Tree_BAO_f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, doub
le * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                double kmag1,
                                double alpha_perp, double alpha_parallel, double sigma2_perp,
double sigma2_para) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_Tree_BAO_f_f_f(kvec1, kvec2, kvec3, los, alpha_per
p, alpha_parallel, sigma2_perp, sigma2_para);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

```

```

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;
}

int integrand_B_NonGaussian_Local_b1_b1_b1(double * xx_in, int ndim, double * ff_out, int
ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Local_b1_b1_b1(kvec1, kvec2, kvec3, lo
s, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_NonGaussian_Local_b1_b1_f(double * xx_in, int ndim, double * ff_out, int
ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/

```

```

double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_Local_b1_b1_f(kvec1, kvec2, kvec3, los
, alpha_perp, alpha_parallel);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_Local_b1_f_f(double * xx_in, int ndim, double * ff_out, int n
comp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_Local_b1_f_f(kvec1, kvec2, kvec3, los,
alpha_perp, alpha_parallel);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * Sl11 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}

```

```

    return 0;

}

int integrand_B_NonGaussian_Local_f_f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Local_f_f_f(kvec1, kvec2, kvec3, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }

    return 0;

}

int integrand_B_NonGaussian_Equilateral_b1_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};

```

```

    kvec2[2]};
    double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_Equilateral_b1_b1_b1(kvec1, kvec2, kve
c3, los, alpha_perp, alpha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_Equilateral_b1_b1_f(double * xx_in, int ndim, double * ff_out
, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1  = 1.0;
    double phi1 = 0.0;
    double mu2  = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu   = - 1.0 + 2.0 * xx_in[1];
    double phi  = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Equilateral_b1_b1_f(kvec1, kvec2, kvec
3, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_Equilateral_b1_f_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_Equilateral_b1_f_f(kvec1, kvec2, kvec3
, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
        ff_out[j] = result * jacobian;

    }
    return 0;
}

```

```

int integrand_B_NonGaussian_Equilateral_f_f_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(

```

```

phi), mu};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_Equilateral_f_f_f(kvec1, kvec2, kvec3,
    los, alpha_perp, alpha_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
    ff_out[j] = result * jacobian;

    }
    return 0;

}

/** PNG from PB decomposed */
int integrand_B_NonGaussian_From_PB_Local_b1_b1_b1_R(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    double R = 10.0;

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b1_b1_R(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
ha_parallel, R);

```

```

/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b1_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b1_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

```

```

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b1_f(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b2_b1(double * xx_in, int ndim, double * ff_

```

```

out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                double kmag1,
                                double alpha_perp, double alpha_parallel
el) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /***/

        /***/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b2_b1(kvec1, kvec2, k
vec3, pvec, los,
                                alpha_perp, alp
ha_parallel);
        /***/

        /***/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /***/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b2_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                double kmag1,
                                double alpha_perp, double alpha_parallel
el) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];

```

```

double phi2 = 0.0;
double mu = -1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b2_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_bK2_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];

```

```

double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /***/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /***/

    /***/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /***/

    /***/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_bK2_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
    /***/

    /***/
    double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N1l1 * S1l1 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_bK2_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};

```

```

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_bK2_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b1f_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/

```

```

        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b1f_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_b1f_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_b1f_f(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);

```

```

double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_ff_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_ff_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

```

```

}
```

```

int integrand_B_NonGaussian_From_PB_Local_b1_ff_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
```

```

el) {
```

```

    /*****/
```

```

    double mu1 = 1.0;
```

```

    double phi1 = 0.0;
```

```

    double mu2 = - 1.0 + 2.0 * xx_in[0];
```

```

    double phi2 = 0.0;
```

```

    double mu = - 1.0 + 2.0 * xx_in[1];
```

```

    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
```

```

    /*****/
```

```

    double kmax = kmax_integ;
```

```

    double kmin = kmin_integ;
```

```

    double dlnk = log(kmax) - log(kmin);
```

```

    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
```

```

    double mu_p = -1.0 + 2.0 * xx_in[4];
```

```

    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];
```

```

    for(int j = 0; j < num_k_bin; j++) {
```

```

        /*****/
```

```

        double kvec1[3] = {0.0, 0.0, kmag1};
```

```

        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
```

```

        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
```

```

kvec2[2]};
```

```

        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
```

```

phi), mu};
```

```

        /*****/
```

```

        /*****/
```

```

        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
```

```

0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
```

```

        /*****/
```

```

        /*****/
```

```

        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_ff_f(kvec1, kvec2, kvec3, pvec, los,
```

```

                                alpha_perp, alpha_parallel);
```

```

    ha_parallel);
```

```

    /*****/
```

```

    /*****/
```

```

    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
```

```

    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
```

```

    double result = N111 * S111 * bispec;
```

```

    /*****/
```

```

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);
```

```

    ff_out[j] = result * jacobian;
```

```

    }
```

```

    return 0;
```

```

}
```

```

int integrand_B_NonGaussian_From_PB_Local_b1_f_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
```

```

double alpha_perp, double alpha_parallel
```

```

1) {
```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_f_b1(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alph
a_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1_f_f(double * xx_in, int ndim, double * ff_ou
t, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                double kmag1,
                                                double alpha_perp, double alpha_parallel
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

```

```

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b1_f_f(kvec1, kvec2, kve
c3, pvec, los,
                                                                    alpha_perp, alpha
_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b2_b1_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

```

```

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b1_b1(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_b1_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

```

```

    /***/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /***/

    /***/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b1_f(kvec1, kvec2, kvec3, pvec, los,
                                                                    alpha_perp, alpha_parallel);
    /***/

    /***/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_b2_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /***/

        /***/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /***/

        /***/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b2_b1(kvec1, kvec2, kvec3, pvec, los,
                                                                    alpha_perp, alpha_parallel);
        /***/
    }
}

```

```

ha_parallel);
    /***/

    /***/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_b2_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /***/

        /***/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b2_f(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /***/

        /***/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /***/

```

```

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_bK2_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_bK2_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b2_bK2_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_bK2_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_b1f_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];

```

```

double phi2 = 0.0;
double mu = -1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b1f_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b2_b1f_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parall
el) {

```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );

```

```

double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /***/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /***/

    /***/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /***/

    /***/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_b1f_f(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /***/

    /***/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_ff_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};

```

```

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_ff_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b2_ff_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

```

```

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_ff_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * Sl11 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_f_b1(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                double kmag1,
                                                double alpha_perp, double alpha_paralle
1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_f_b1(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alph
a_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E

```

```

LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b2_f_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /***/

        /***/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /***/

        /***/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b2_f_f(kvec1, kvec2, kve
c3, pvec, los,
alpha_perp, alpha_parallel);
        /***/

        /***/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /***/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
}

```

```

    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_b1_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b1_b1(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_b1_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1,
```

```

double alpha_perp, double alpha_parallel

el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b1_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_b2_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel

el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

```

```

    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /**/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /**/

        /**/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /**/

        /**/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b2_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /**/

        /**/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /**/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_bK2_b2_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

```

```

    /**/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /**/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

```

```

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b2_f(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_bK2_b1(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

```

```

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_bK2_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_NonGaussian_From_PB_Local_bK2_bK2_f(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_bK2_f(kvec1, kvec2,
kvec3, pvec, los,

```

alpha_perp, alp

```

ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_b1f_b1(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b1f_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
pha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

```

```

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_b1f_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_b1f_f(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_bK2_ff_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_ff_b1(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);
        ff_out[j] = result * jacobian;

    }
    return 0;
}

int integrand_B_NonGaussian_From_PB_Local_bK2_ff_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;

```

```

double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_ff_f(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_bK2_f_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_paralle
1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);

```

```

double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /***/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /***/

    /***/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /***/

    /***/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_f_b1(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alph
a_parallel);
    /***/

    /***/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_bK2_f_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

```

```

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};

```

```

double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_bK2_f_f(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alpha
_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b1f_b1_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};

```

```

/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_b1_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_b1_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_b1_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

```

```

/*****/
double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_b2_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_b2_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

```

```

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_b2_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*******/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*******/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*******/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*******/

        /*******/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*******/

        /*******/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_b2_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*******/

        /*******/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*******/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_bK2_b1(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel

```

```

el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_bK2_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b1f_bK2_f(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

```

```

    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_blf_bK2_f(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double Nl1l = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl1l = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl1l * Sl1l * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_blf_blf_b1(double * xx_in, int ndim, double * f
f_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

```

```

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_b1f_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_b1f_f(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

```

```

    /***/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /***/

    /***/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_blf_blf_f(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alp
ha_parallel);
    /***/

    /***/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(ELL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * S111 * bispec;
    /***/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_blf_ff_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /***/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /***/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /***/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] - kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(phi), mu};
        /***/

        /***/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /***/

        /***/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_blf_ff_b1(kvec1, kvec2, kvec3, pvec, los, alpha_perp, alp
ha_parallel);
        /***/

    }

}

```

```

ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_ff_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_ff_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

```

```

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_b1f_f_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_paralle
1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_f_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alph
a_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_b1f_f_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_b1f_f_f(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alpha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_b1_b1(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

    /*****/

```

```

double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b1_b1(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_b1_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;

```

```

double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b1_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_ff_b2_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};

```

```

double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b2_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_ff_b2_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.

```

```

0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b2_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_bK2_b1(double * xx_in, int ndim, double * ff
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parall
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_bK2_b1(kvec1, kvec2,
kvec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
        /*****/

```

```

    /*****/
    double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N1l1 * S1l1 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_bK2_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_bK2_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

```

```

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_b1f_b1(double * xx_in, int ndim, double * ff_
_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b1f_b1(kvec1, kvec2,
kvec3, pvec, los,
alpha_perp, al
pha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_b1f_f(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
```

```

double alpha_perp, double alpha_parallel

el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_b1f_f(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_ff_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel

el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];

```

```

    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_ff_b1(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N1l1 * S1l1 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_ff_ff_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

```

```

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_ff_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_f_b1(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_paralle
l) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};

```

```

/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_f_b1(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alph
a_parallel);
/*****/

/*****/
double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N111 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_ff_f_f(double * xx_in, int ndim, double * ff_ou
t, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_ff_f_f(kvec1, kvec2, kve

```

c3, pvec, los,

alpha_perp, alpha

_parallel);

/******/

/******/

double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);

double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);

double result = N111 * S111 * bispec;

/******/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;

ff_out[j] = result * jacobian;

}

return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_b1_b1(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,

double kmag1,

double alpha_perp, double alpha_parallel

el) {

/******/

double mu1 = 1.0;

double phi1 = 0.0;

double mu2 = - 1.0 + 2.0 * xx_in[0];

double phi2 = 0.0;

double mu = - 1.0 + 2.0 * xx_in[1];

double phi = 0.0 + 2.0 * M_PI * xx_in[2];

/******/

double kmax = kmax_integ;

double kmin = kmin_integ;

double dlnk = log(kmax) - log(kmin);

double pmag = exp(log(kmin) + dlnk * xx_in[3]);

double mu_p = -1.0 + 2.0 * xx_in[4];

double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/******/

double kvec1[3] = {0.0, 0.0, kmag1};

double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};

double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};

double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};

/******/

/******/

double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};

/******/

/******/

double bispec = Bispectrum_NonGaussian_From_PB_Local_f_b1_b1(kvec1, kvec2, kv
ec3, pvec, los,

alpha_perp, alp

ha_parallel);

/******/

/******/

double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);

double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);

```

double result = Nl11 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_b1_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_f_b1_f(kvec1, kvec2, kve
c3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * S111 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

```

```

}

int integrand_B_NonGaussian_From_PB_Local_f_b2_b1(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_f_b2_b1(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = N111 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_b2_f(double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL, double kmag1, double alpha_perp, double alpha_parallel) {

    /*****/

```

```

double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_f_b2_f(kvec1, kvec2, kve
c3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_bK2_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;

```

```

double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_f_bK2_b1(kvec1, kvec2, k
vec3, pvec, los,
                                                                    alpha_perp, al
pha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_NonGaussian_From_PB_Local_f_bK2_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

```

```

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};

```

```

    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
    double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_f_bK2_f(kvec1, kvec2, kv
ec3, pvec, los,
                                                                    alpha_perp, alp
ha_parallel);
    /*****/

    /*****/
    double N111 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = N111 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_b1f_b1(double * xx_in, int ndim, double * ff_
out, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
                                                                    double kmag1,
                                                                    double alpha_perp, double alpha_parallel) {

    /*****/
    double mu1  = 1.0;
    double phi1 = 0.0;
    double mu2  = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu   = - 1.0 + 2.0 * xx_in[1];
    double phi  = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};

```

```

/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_f_blf_b1(kvec1, kvec2, k
vec3, pvec, los,
alpha_perp, al
pha_parallel);
/*****/

/*****/
double N1l1 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S1l1 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = N1l1 * S1l1 * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_blf_f(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_f_blf_f(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

```

```

/*****/
double Nl1l = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl1l = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl1l * Sl1l * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

}
return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_ff_b1(double * xx_in, int ndim, double * ff_o
ut, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

/*****/
double mu1 = 1.0;
double phi1 = 0.0;
double mu2 = - 1.0 + 2.0 * xx_in[0];
double phi2 = 0.0;
double mu = - 1.0 + 2.0 * xx_in[1];
double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

/*****/
double kvec1[3] = {0.0, 0.0, kmag1};
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/

/*****/
double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
/*****/

/*****/
double bispec = Bispectrum_NonGaussian_From_PB_Local_f_ff_b1(kvec1, kvec2, kv
ec3, pvec, los,
alpha_perp, alp
ha_parallel);
/*****/

/*****/
double Nl1l = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double Sl1l = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl1l * Sl1l * bispec;
/*****/

double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
ff_out[j] = result * jacobian;

```

```

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_ff_f(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
el) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_f_ff_f(kvec1, kvec2, kve
c3, pvec, los,
alpha_perp, alp
ha_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * S111 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_f_b1(double * xx_in, int ndim, double * ff_out,
int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
```

```

double alpha_perp, double alpha_parallel

1) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    double kmax = kmax_integ;
    double kmin = kmin_integ;
    double dlnk = log(kmax) - log(kmin);
    double pmag = exp( log(kmin) + dlnk * xx_in[3] );
    double mu_p = -1.0 + 2.0 * xx_in[4];
    double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};
        double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
        double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
        double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
        /*****/

        /*****/
        double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
        /*****/

        /*****/
        double bispec = Bispectrum_NonGaussian_From_PB_Local_f_f_b1(kvec1, kvec2, kve
c3, pvec, los,
alpha_perp, alph
a_parallel);
        /*****/

        /*****/
        double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
        double Sl11 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
        double result = Nl11 * Sl11 * bispec;
        /*****/

        double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
        ff_out[j] = result * jacobian;

    }
    return 0;

}

int integrand_B_NonGaussian_From_PB_Local_f_f_f(double * xx_in, int ndim, double * ff_out
, int ncomp, double * kbin, int num_k_bin, int ell1, int ell2, int ELL,
double kmag1,
double alpha_perp, double alpha_parallel
) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = - 1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = - 1.0 + 2.0 * xx_in[1];

```

```

double phi = 0.0 + 2.0 * M_PI * xx_in[2];
/*****/

double kmax = kmax_integ;
double kmin = kmin_integ;
double dlnk = log(kmax) - log(kmin);
double pmag = exp( log(kmin) + dlnk * xx_in[3] );
double mu_p = -1.0 + 2.0 * xx_in[4];
double phi_p = 0.0 + 2.0 * M_PI * xx_in[5];

for(int j = 0; j < num_k_bin; j++) {

    /*****/
    double kvec1[3] = {0.0, 0.0, kmag1};
    double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
    double kvec3[3] = {-kvec1[0] - kvec2[0], -kvec1[1] - kvec2[1], -kvec1[2] -
kvec2[2]};
    double los[3] = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
    /*****/

    /*****/
    double pvec[3] = {pmag * sqrt(1.0 - mu_p * mu_p) * cos(phi_p), pmag * sqrt(1.
0 - mu_p * mu_p) * sin(phi_p), pmag * mu_p};
    /*****/

    /*****/
    double bispec = Bispectrum_NonGaussian_From_PB_Local_f_f_f(kvec1, kvec2, kvec
3, pvec, los,
alpha_perp, alpha
_parallel);
    /*****/

    /*****/
    double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
    double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
    double result = Nl11 * S111 * bispec;
    /*****/

    double jacobian = dlnk * pow(pmag,3) / (2.0 * M_PI * M_PI);;
    ff_out[j] = result * jacobian;

}
return 0;

}

```

```

int integrand_B_Tree_BAO_Template(
    double * xx_in, int ndim, double * ff_out, int ncomp, double * kbin, int num_k_bi
n, int ell1, int ell2, int ELL, double kmag1,
    double alpha_perp, double alpha_parallel, double sigma2_perp, double sigma
a2_para,
    char * parameters) {

    /*****/
    double mu1 = 1.0;
    double phi1 = 0.0;
    double mu2 = -1.0 + 2.0 * xx_in[0];
    double phi2 = 0.0;
    double mu = -1.0 + 2.0 * xx_in[1];
    double phi = 0.0 + 2.0 * M_PI * xx_in[2];
    /*****/

    for(int j = 0; j < num_k_bin; j++) {

        /*****/
        double kvec1[3] = {0.0, 0.0, kmag1};

```

```
double kvec2[3] = {kbin[j] * sqrt(1.0 - mu2 * mu2), 0.0, kbin[j] * mu2};
double kvec3[3] = {- kvec1[0] - kvec2[0], - kvec1[1] - kvec2[1], - kvec1[2] -
kvec2[2]};
double los[3]    = {sqrt(1.0 - mu * mu) * cos(phi), sqrt(1.0 - mu * mu) * sin(
phi), mu};
/*****/
double bispec = Bispectrum_Tree_BAO_Template(kvec1, kvec2, kvec3, los, alpha_
perp, alpha_parallel, sigma2_perp, sigma2_para, parameters);
/*****/

/*****/
double Nl11 = (2.0*double(ell1)+1.0) * (2.0*double(ell2)+1.0) * (2.0*double(E
LL)+1.0);
double S111 = calcYYY(mu1, phi1, mu2, phi2, mu, phi, ell1, ell2, ELL);
double result = Nl11 * S111 * bispec;
/*****/

double jacobian = (2.0 / 2.0) * (2.0 / 2.0) * (2.0 * M_PI / (2.0 * M_PI));
ff_out[j] = result * jacobian;

}

return 0;

}
```

#endif